CFENet: Contextual Feature Enhancement Network for Tiny Object Detection in Aerial Images

Heng Hu, Si-Bao Chen, *Member, IEEE*, and Jin Tang

***Abstract*— With the development of deep learning techniques and object detectors, the performance of object detection has been rapidly improved. However, since tiny objects contain only a small number of pixels and lack appearance information, this creates difficulties for detector recognition. Although existing research has improved detection performance by fusing different feature layers to enhance feature information of objects, this also leads to the problem of mixed feature information, especially for tiny objects where features are easily covered, which exacerbates the difficulty of recognition. To solve the above problems, we propose a contextual feature enhancement network (CFENet), which is an efficient framework built on anchor-based object detectors. In CFENet, to effectively utilize contextual information around an object to enhance the detection of tiny objects, we use poolFormer to build a backbone to extract object features. To alleviate the feature blending problem caused by feature fusion, we propose a feature suppression module (FSM) that effectively suppresses background information and redundant features to enhance tiny object features. In addition, we utilize the improved Gaussian Wasserstein distance loss to modify the loss function to obtain high-quality bounding boxes, and we further manipulate the shallow feature layer of the output and then add a detection head to enhance the detection of tiny objects. We have conducted extensive experiments on the public datasets AI-TOD, VisDrone, and DOTA to demonstrate the effectiveness of our approach.**

***Index Terms*— Feature pyramid network, feature suppression module (FSM), improved Gaussian Wasserstein distance (IGWD) loss, tiny object detection.**

I. Introduction

With the development of deep learning techniques, the performance of object detection has rapidly improved in a short period of time. In contrast to small, medium, and large objects defined in the MS COCO benchmark [1], the AI-TOD benchmark [2] defines tiny objects as those with dimensions smaller than 16 × 16 pixels. However, compared with the detection of larger objects, small objects themselves have fewer pixels in their extremely small size, weaker feature rendering ability, and their appearance information is extremely limited, resulting in less spatial information, and tiny objects are easily confused with the background, increasing background interference [3], [4]. Due to the limitation of spatial resolution, it is difficult for conventional object detectors to effectively capture the features required for accurate classification and localization of tiny objects, resulting in a large number of failures in detecting tiny objects [2], [5], [6], [7].

To get better performance on object detection, researchers have been exploring various approaches, including data augmentation, scale-aware, feature fusion, and so on. In general, deep features contain more semantic information about large objects, while small objects are characterized in the shallow layer with large feature resolution [8], [9]. In order to effectively utilize feature information at different scales, shallow feature maps detect tiny objects and deep feature layers detect large objects, HyperNet [10] focuses on effectively organizing and utilizing information from different layers of the backbone network. By splicing and compressing coarse-to-fine features, HyperNet constructs a hyperfeature representation that retains crucial details for detecting small objects. This method allows for the preservation of spatial information across multiple scales, which is essential for accurately localizing tiny objects amidst cluttered backgrounds.

However, in traditional convolutional neural networks (CNNs), a cascade structure is usually used to extract features, but this structure may lead to a gradual enhancement of semantic information but a gradual loss of detailed information as the depth of the network increases. The limitations of the cascade method lead to the lack of semantic information and detailed information in the shallow feature and deep feature layers, respectively. Therefore, to solve the problem of missing information on the multiscale feature layer, researchers investigated feature fusion methods to further improve the detection performance by fusing feature layers of different scales so that the shallow and deep feature layers have semantic information and detail information, respectively. For example, the core idea of FPN [11] is to obtain multiscale feature representations by constructing feature pyramids at different network levels. Specifically, FPN constructs the feature pyramid by top-down recursion, from the high level to the low level of the network, and obtains high-resolution feature maps by upsampling and merging operations. In this way, rich semantic and detailed information can be obtained at different scales. QueryDet [12] introduces a cascading query strategy to optimize the utilization of underlying features during the detection process.

Received 4 June 2024; revised 17 November 2024 and 17 January 2025; accepted 27 February 2025. Date of publication 31 March 2025; date of current version 28 April 2025. This work was supported in part by the NSFC Key Project of Joint Fund for Enterprise Innovation and Development under Grant U20B2068 and in part by the National Natural Science Foundation of China under Grant 62106006 and Grant 61976004. *(Corresponding author: Si-Bao Chen.)*

The authors are with the MOE Key Laboratory of ICSP, IMIS Laboratory of Anhui Province, Anhui Provincial Key Laboratory of Multimodal Cognitive Computation, Zenmorn-AHU AI Joint Laboratory, School of Computer Science and Technology, Anhui University, Hefei 230601, China (e-mail: e22101010@stu.ahu.edu.cn; sbchen@ahu.edu.cn; tangjin@ahu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3556469

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:07:24 UTC from IEEE Xplore. Restrictions apply.

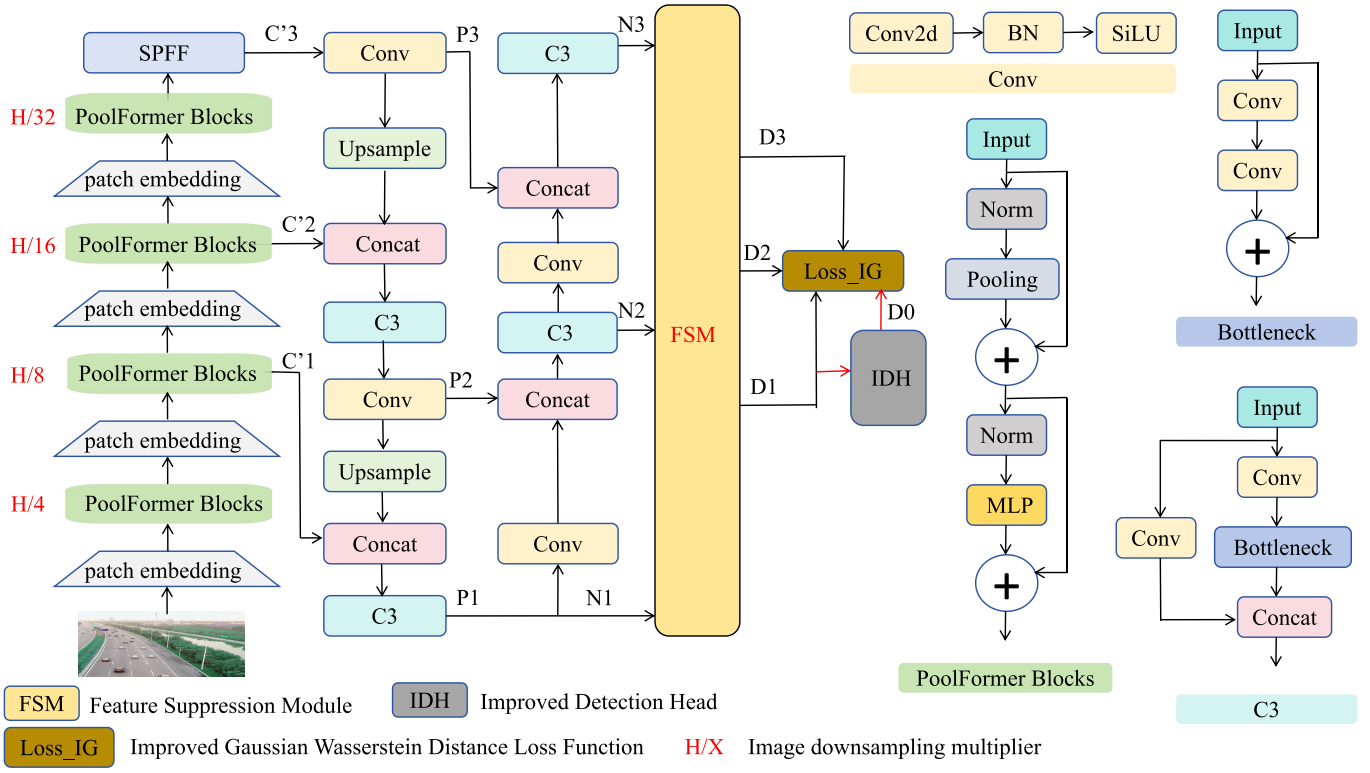


Fig. 1. Model architecture of CFENet. First, we build a backbone mesh based on poolFormer to better utilize the contextual information around the object to extract feature information and assist in object detection. Then we extract the feature layers with downsampling multiples of 8, 16, and 32 from the backbone network as the base feature layer. Then PAN feature fusion is performed on the base feature layer. To alleviate the feature mixing problem, we embed the FSM mechanism after PAN. To further enhance the detection performance of tiny objects, we add additional detection heads to the output shallow feature layer. Finally, when calculating the loss, we use IGWD instead of the original loss function.

By strategically selecting and processing regions of interest (RoIs) based on cascading queries, QueryDet minimizes redundant computations, particularly on high-resolution feature maps. This strategy enables efficient detection of small objects while reducing computational overhead.

The introduction of feature fusion does solve the problem of insufficient shallow semantic information and lack of deep detail information, thus improving the detection performance. However, feature fusion also brings a potential problem: feature blending. When object features of different scales are fused to the same feature map, it may lead to feature scale confusion, thus affecting the detection performance. To address this challenge, we propose the feature suppression module (FSM), this module aims to mitigate the feature scale confusion problem by suppressing feature redundant features and complex background information. This in turn helps to maintain the discriminative power of features at different scales and improve the detection performance. Importantly, this FSM can be seamlessly integrated into any feature fusion-based detector, making it a versatile solution for enhanced object detection.

In addition, intersection over union (IoU) [13] is a metric used to measure the degree of overlap between the model-predicted bounding box and the ground true bounding box. However, for tiny objects, IoU is more sensitive because the bounding boxes of tiny objects are usually small and a slight positional shift may lead to a large IoU change, which is analyzed in both dot distance (DotD) [14] and normalized Wasserstein distance (NWD) [7]. Although DotD and NWD are smoother compared to IoU, they do not have scale invariance. This situation may seriously affect the performance and accuracy of object detection. To mitigate the sensitivity of the IoU series metrics to the positional offset of tiny objects to solve the problem that NWD does not have scale invariance, and realize more accurate tiny object detection, we design the improved Gaussian Wasserstein distance (IGWD) loss as the regression loss of the model.

Meanwhile, to achieve more effective and accurate detection of small objects, we constructed a backbone network using the poolFormer [15] module, which can effectively utilize the contextual information around the objects to assist in object detection. In addition, we further manipulate the output shallow feature layer and then add a detection head to improve the detection performance for tiny objects. In summary, these improvements allow the model to be more applicable to remote sensing tiny object detection, as shown in Fig. 1, which demonstrates our proposed contextual feature enhancement network (CFENet) model. In summary, the main contributions are as follows.

1) We propose a novel and efficient anchor box object detection network, CFENet, which achieves state-of-the-art performance on public object datasets AI-TOD, VisDrone, and DOTA.
2) We propose the FSM, which effectively mitigates feature mixing problems caused by feature fusion and improves the detection accuracy of tiny objects.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:07:24 UTC from IEEE Xplore. Restrictions apply.

3) We design an IGWD loss function that is more effective in detecting tiny objects.
4) We introduce the poolFormer module to build a new backbone, which effectively utilizes contextual information around the object. An improved detection head is also proposed to enhance the detection of the object.

II. Related Work

A. Feature Fusion Methods

Cascaded CNN is a common multiscale feature extraction method that generates feature maps of different resolution sizes by stacking multiple CNN models and passing features between them. In cascaded CNNs, shallow feature maps are usually rich in detailed information, while deep feature maps are rich in more advanced semantic information. However, due to the presence of subsampling layers, deep feature maps may result in the disappearance of the response of small objects due to the loss of detailed information about the objects after multiple downsampling, making it difficult to accurately detect small objects. To overcome this difficulty, researchers have used feature fusion techniques to fuse feature layers at different scales to obtain a better characterization of object features. Path aggregation network (PANet) [16] introduced bidirectional paths to enrich the feature hierarchy. By establishing bidirectional paths, PANet could facilitate the aggregation of bottom-up and top-down path information. Bidirectional feature pyramid network (BiFPN) [17] entered into the bidirectional information flow and realized the bidirectional transfer of information by introducing bidirectional connections in the feature pyramid network. Specifically, BiFPN introduced an adaptive fusion mechanism at each layer of the feature pyramid network that automatically learned and fused feature information at different scales. This bidirectional information flow enabled the network to better capture the correlation between feature maps and improved the performance of object detection. FFCA-YOLO [18] proposed an efficient feature enhancement, fusion, and context-aware object detector, which effectively suppressed the background interference and enhanced the object features. Ma et al. [19] similarly by manipulating the extracted features, the object features were effectively enhanced. Recursive feature pyramid network (RecursiveFPN) [20] enhanced the representation of multiscale features by recursively constructing feature pyramids. Unlike traditional feature pyramid networks, RecursiveFPN not only constructed the feature pyramid once in the network but generated multiple levels of feature pyramids by recursively applying the feature pyramid construction process. This recursive feature pyramid construction approach better-captured feature information at different scales.

B. Loss Functions

Loss functions in object detection are often used to measure the discrepancy between the model prediction and the real object to adjust the model parameters to minimize this discrepancy by optimizing the algorithm. Unitbox proposes an IoU loss that regresses the four boundaries of the bounding box as a whole, rather than regressing each boundary individually. However, the IoU loss function may cause the gradient to vanish when there is no overlap between the bounding boxes, thus affecting the training and performance of the model. Thus, Rezatofighi et al. [21] introduced the generalized IoU (GIoU) in their work to address the problem of vanishing gradients generated by the IoU loss function when the regions between the regression and ground true boxes did not overlap. Specifically, the GIoU loss took into account not only the size of the overlap region, but also the size of the outer rectangles of the predicted bounding box and the ground true bounding box, which allowed the loss function to provide meaningful gradient signals even when there was no overlap, which helped to better guide the training of the model. Distance IoU (DIoU) and complete IoU (CIoU) were proposed by Zheng et al. [22] to further improve the bounding box regression loss in object detection. These methods aimed to address the problem of slow convergence of GIoU in some cases and to penalize the location and shape of the bounding box more effectively in the optimization process. The embedding IoU (EIoU) loss proposed by Zhang et al. [23] was a novel loss function for the bounding box regression task, which aimed to characterize the geometric features of the object box more efficiently and improved the accuracy and convergence speed of the regression results. By representing bounding boxes as embedding vectors and operating in the embedding space, the EIoU loss was able to more accurately measure the similarity between bounding boxes, thus better-guiding model training and optimization. Gevorgyan [24] proposed the SIoU (Directional IoU) loss function as a novel loss function for object detection, which solved the problem of mismatch between the expected ground truth bounding box and the predicted bounding box direction ignored in the traditional loss function by considering the angle between the expected regression vectors.

C. Context-Based Detection

When an object is small or has weak features, relying on the features of the object itself alone may not be sufficient for accurate detection. At this point, utilizing the background and environmental information around the object can provide additional context to help the model better understand the image content and improve the accuracy of the detection. Chen et al. [25] utilized the representations of context regions, which encompassed the proposal patches, to facilitate subsequent recognition tasks. This approach enabled the model to consider the context surrounding the object proposals, improving recognition accuracy. PyramidBox [26] leveraged contextual cues comprehensively to detect small and blurry faces that might blend into the background. By incorporating contextual information, PyramidBox improved the ability to distinguish small faces from the surrounding environment. FS-SSD [27] exploited implicit spatial context information to improve object detection performance, particularly for objects with low confidence scores. SINet [28] introduced a context-aware RoI pooling layer that helps to improve the detection of small objects by retaining contextual information while pooling features of the region of interest. By incorporating global contextual information, particularly for heavily

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:07:24 UTC from IEEE Xplore. Restrictions apply.

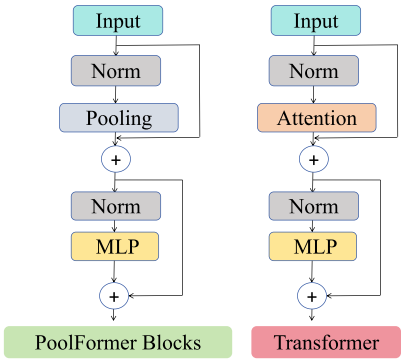


Fig. 2. Model architecture of transformer and poolFormer. The difference between poolFormer and a transformer is that a transformer uses a module based on the attention mechanism and a poolFormer uses a pooling module instead of an attention module.

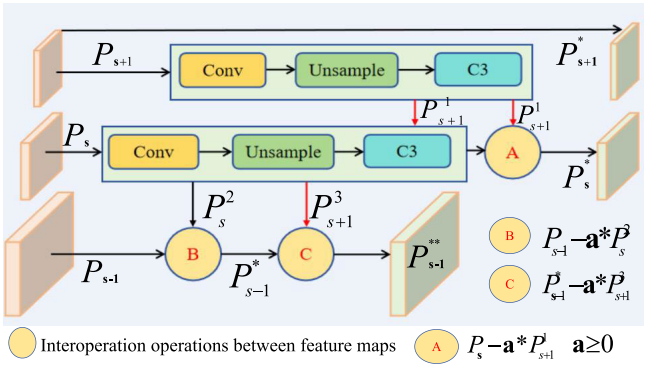


Fig. 3. Model architecture of FSM. P_{s+1}, P_s, and P_{s-1} are the three different scale feature layers output after feature fusion, and P^*_{s+1}, P^*_s, and $P^{**}_{s_1}$ feature layers are the corresponding different scale feature layers output after FSM mechanism. Note that P_{s+1} and P^*_{s+1} are the same feature layer.

occluded objects, IONet [29] enhanced the detection performance, especially for small and heavily occluded objects. Ma et al. [4] and Zhang et al. [30] similarly fully and effectively utilized the contextual information of the object to assist the object in detection and improve the detection performance.

III. Methodology

A. Preliminary

Transformer [31] model is able to capture the associations between different regions in an image through its self-attention mechanism, which provides richer contextual information. This mechanism enables the transformer to better understand the context around the object when dealing with small objects, thus improving the detection performance of tiny objects. CNN-based detectors capture image features through their local receptive fields, which can lead to a gradual loss of information about features of small objects in the cascade approach to extraction. To overcome the limitations of a single model, the researchers tried to combine a transformer with CNN. This approach combines the local feature-capturing ability of CNN and the global context encoding ability of the transformer, which enables the model to better utilize the correlation information between different regions in the image, thus improving the performance of small object detection. The transformer encoder, shown in Fig. 2, consists of two components. One is an attention module for mixing information between tokens. The other component contains the remaining modules such as channel MLP and residual connections.

To effectively utilize the contextual information around the object to assist the detection of tiny objects, the model backbone of this article is constructed by the poolFormer module, as shown in Fig. 2. The poolFormer module was proposed by Yu et al. [15] who replaced the attention module in the transformer using a very simple nonparametric operator space, which outperforms the transformer and its variants.

B. Feature Suppression Module

Although feature fusion improves the performance of object detection, it can lead to the feature mixing problem due to the fusion of features from other feature layers in the same feature layer, especially for tiny objects, whose features are easy to be covered by large object features and background information, which exacerbates the difficulty of recognition. To alleviate the feature mixing problem, we propose an FSM mechanism to suppress background information and redundant features and enhance tiny object features.

As shown in Fig. 3, in FSM, we assume that the three neighboring feature layers are denoted as P_{s+1}, P_s, and P_{s-1}, and all three are feature layers formed on the basis of feature fusion (e.g., FPN or PAN). Among them, the shallow feature layer P_{s-1} and the intermediate feature layer P_s are responsible for detecting small objects, and the deep feature layer P_{s+1} is responsible for detecting large objects. However, due to the characteristics of feature fusion methods (e.g., top-down FPN), the P_s layer contains the feature information of the P_{s+1} layer, resulting in that the small object feature information of the Ps feature layer is easily overwritten by the large object features of the P_{s+1} feature layer. The P_{s-1} layer contains the feature information of the P_s layer and the P_{s+1} layer, which also tends to cause the small object features in the P_{s-1} feature layer to be easily covered by the small object features in the P_{s+1} feature layer. Therefore, to enhance the features of small objects and alleviate the feature mixing problem in the P_s layer and the P_{s-1} layer, it is necessary to embed our proposed FSM into the detector model.

As shown in Fig. 3, to suppress the feature information from the P_{s+1} layer, the following operations are performed for the P_s feature layer: first, the P_{s+1} feature map passes through a 1×1 convolution layer with a step size of 1, so that the number of channels of the convolved feature map is the same as the number of channels of the P_s feature map. Then an upsampling module with a step size of 2 is used to make the size of the upsampled feature map the same as the size of the P_s layer feature map. Then the newly obtained feature map is subjected to C_3 operations to extract more feature information to obtain the P^1_{s+1} layer feature map. Depending on the network model, C_3 modules can be replaced with components from the corresponding model.

$$P^1_{s+1} = c_3(u(c_{1\times1}(P_{s+1}))) \tag{1}$$

where $c_{1\times1}$ denotes a 1×1 convolution with step 1, u denotes an upsampling with step 2, and c_3 denotes the C_3 module in the YOLOv5 [32] model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:07:24 UTC from IEEE Xplore. Restrictions apply.

Finally, the P_s layer feature map is used to perform an A operation with the P_{s+1}^1 layer feature map to obtain the final midlayer output feature map P_s^*. The A operation is calculated as follows:

$$P_s^* = P_s - a \times P_{s+1}^1 \tag{2}$$

where a is a constant greater than or equal to zero. After the operation as above, the P_s feature layer successfully suppresses the feature information from the P_{s+1} feature layer and enhances the features of the object in this layer.

To suppress the feature information from the P_{s+1} feature layer and the P_s feature layer, the following operation is performed for the P_{s-1} feature layer.

The P_{s+1}^1 layer feature map passes through a 1×1 convolution layer with a step size of 1, so that the number of channels of the feature map after convolution is the same as the number of channels of the P_{s-1} layer feature map, and then passes through an upsampling module with a step size of 2, so that the size of the upsampled feature map is the same as the size of the P_{s-1} layer feature map, and then performs C_3 operations on the newly obtained feature map to obtain the final P_{s+1}^3 layer feature map

$$P_{s+1}^3 = c_3\left(u\left(c_{1\times1}\left(P_{s+1}^1\right)\right)\right) \tag{3}$$

where $c_{1\times1}$ denotes a 1×1 convolution with step 1, u denotes an upsampling with step 2, and c_3 denotes the C_3 module in the YOLOv5 model.

Then the P_s layer is operated as follows: first, the P_s feature map passes through a 1×1 convolution layer with a step size of 1, so that the number of channels of the feature map after convolution is the same as the number of channels of the feature map of the P_{s-1} layer, and then it passes through an upsampling module with a step size of 2, so that the size of the feature map after upsampling is the same as the size of the feature map of the P_{s-1} layer, and then it performs C_3 operation three times on the new feature map to get the final P_s^2 layer feature map

$$P_s^2 = c_3(u(c_{1\times1}(P_s))) \tag{4}$$

where $c_{1\times1}$ denotes a 1×1 convolution with step 1, u denotes an upsampling with step 2, and c_3 denotes the C_3 module in the YOLOv5 model. The P_{s-1} layer feature map is then used with the newly obtained P_s^2 layer feature map and P_{s+1}^3 layer feature map to perform B and C operations to obtain the final shallow output feature map P_{s-1}^{**}. The B and C operations are computed as follows:

$$P_{s-1}^{**} = P_{s-1} - a \times P_s^2 - a \times P_{s+1}^3 \tag{5}$$

where a is a constant greater than or equal to zero. After the operation as above, the P_{s-1} layer successfully suppresses the feature information from the P_s and P_{s+1} layers and enhances the feature information of the object in this layer.

In conclusion, we have not only solved the feature mixing problem caused by feature fusion by FSM, but suppressed the background information and enhanced the object features. At the same time, it retains the advantages of feature fusion with semantic information in the shallow layer and detail information in the deep layer, which further improves detection accuracy.

C. Loss Function

Regression loss The detector model has limitations in using the loss of IoU series metrics alone, because IoU series metrics are very sensitive to slight shifts in the position of tiny objects, and the use of IoU loss leads to many false positive and false negative problems. Therefore, to improve the detection performance of tiny objects and alleviate the sensitivity of the IoU series metrics to the positional offset of tiny objects, we improve the GWD [33], and then combine it with the IoU loss to design a new IGWD loss that is more suitable for tiny objects. Specifically, we first convert the bounding box into a Gaussian distribution, then solve the distance between two Gaussian distributions and the area of two bounding boxes, then mathematically process the distance and area to get the new metric IGWD, then normalize IGWD, and finally weight and sum it with the IoU series of metrics as the final loss function IGWD loss.

Here we rephrase the main idea in the recent work GWD [33], which converts an arbitrary-oriented bounding box $B(x_t, y_t, w_t, h_t, \theta)$ into a 2-D Gaussian density function $N(\mu, \Sigma)$. Specifically, the conversion is

$$\begin{aligned}
\boldsymbol{\mu} &= (x_t, y_t)^\top \\
\boldsymbol{\Sigma}^{1/2} &= \mathbf{R}\lambda\mathbf{R}^\top = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \\
&\quad \times \begin{pmatrix} \frac{w_t}{2} & 0 \\ 0 & \frac{h_t}{2} \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \\
&= \begin{pmatrix} \frac{w_t}{2}\cos^2\theta + \frac{h_t}{2}\sin^2\theta & \frac{w_t - h_t}{2}\cos\theta\sin\theta \\ \frac{w_t - h_t}{2}\cos\theta\sin\theta & \frac{w_t}{2}\sin^2\theta + \frac{h_t}{2}\cos^2\theta \end{pmatrix}
\end{aligned} \tag{6}$$

where $\mathbf{R}$ represents the rotation matrix and λ represents the diagonal matrix of eigenvalues. Because the boxes are horizontal ($\theta = 0°$), (6) can be further simplified

$$\mu = \begin{bmatrix} cx_t \\ cy_t \end{bmatrix}, \Sigma = \begin{bmatrix} \frac{w_t^2}{4} & 0 \\ 0 & \frac{h_t^2}{4} \end{bmatrix}. \tag{7}$$

Suppose there are two Gaussian distributions, denoted as $\mathbf{X}_p \sim \mathcal{N}_p(\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$ and $\mathbf{X}_t \sim \mathcal{N}_t(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$.

1) Wasserstein Distance: The Wasserstein distance two probability measures are expressed as follows:

$$\begin{aligned}
\mathbf{D}_w\left(\mathcal{N}_p, \mathcal{N}_t\right)^2 &= \mathbf{Tr}\left(\boldsymbol{\Sigma}_p + \boldsymbol{\Sigma}_t - 2\left(\boldsymbol{\Sigma}_p^{1/2}\boldsymbol{\Sigma}_t\boldsymbol{\Sigma}_p^{1/2}\right)^{1/2}\right) \\
&\quad + \|\boldsymbol{\mu}_p - \boldsymbol{\mu}_t\|_2^2.
\end{aligned} \tag{8}$$

Because all the boxes are horizontal ($\theta = 0°$), (8) can be further simplified

$$\mathbf{D}_w^h\left(\mathcal{N}_p, \mathcal{N}_t\right)^2 = \|\mathbf{m}_1 - \mathbf{m}_2\|_2^2 + \|\boldsymbol{\Sigma}_1^{1/2} - \boldsymbol{\Sigma}_2^{1/2}\|_F^2 \tag{9}$$

where $\|\cdot\|_F$ is the Frobenius norm.

Furthermore, for Gaussian distributions $\mathcal{N}_p$ and $\mathcal{N}_t$ which are modeled from bounding boxes $O_p = (x_p, y_p, w_p, h_p)$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:07:24 UTC from IEEE Xplore. Restrictions apply.

and $O_t = (x_t, y_t, w_t, h_t)$, (9) can be further simplified as follows:

$$\begin{aligned} W_2^2(\mathcal{N}_p, \mathcal{N}_t) &= \left\| \left(\left[x_p, y_p, \frac{w_p}{2}, \frac{h_p}{2} \right]^{\mathrm{T}}, \left[x_t, y_t, \frac{w_t}{2} \frac{h_t}{2} \right]^{\mathrm{T}} \right) \right\|_2^2 \\ &= l_2\text{-norm} \left(\left[x_p, y_p, \frac{w_p}{2}, \frac{h_p}{2} \right]^{\top}, \left[x_t, y_t, \frac{w_t}{2}, \frac{h_t}{2} \right]^{\top} \right) \\ &= \left(x_p - x_t \right)^2 + \left(y_p - y_t \right)^2 \\ &\quad + \frac{\left(w_p - w_t \right)^2 + \left(h_p - h_t \right)^2}{4}. \end{aligned} \tag{10}$$

Sum of the areas of the two bounding boxes O_p and O_t are calculated as follows:

$$S(O_p, O_t) = w_p \times h_p + w_t \times h_t. \tag{11}$$

The IGWD-based loss can be expressed as follows:

$$L_{\mathrm{IG}} = \frac{W_2^2(\mathcal{N}_p, \mathcal{N}_t)}{S(O_p, O_t)}. \tag{12}$$

Note that L_{IG} cannot be directly used as loss and needs to be normalized. Therefore, we normalize it to obtain the new loss L_{NIG}

$$L_{\mathrm{NIG}} = e^{-\lambda \times \sqrt{L_{\mathrm{IG}}}} \tag{13}$$

or

$$L_{\mathrm{NIG}} = \frac{1}{1 + \lambda \times L_{\mathrm{IG}}} \tag{14}$$

where λ is a constant greater than zero.

The final loss function is expressed as follows:

$$L_{\mathrm{NIG}^*} = b \times L_{\mathrm{NIG}} + c \times L_{\mathrm{IoU}^*} \tag{15}$$

where constant b and c is greater than or equal to zero and $b + c = 1$, IoU* denotes a family of IoU metrics.

D. Improved Detection Head

To enhance the detection of small objects, we further manipulate the output shallow feature layer D_1 to obtain the feature layer D_0, and then add a detection head on the feature layer D_0, as shown in Fig. 4. This is because the backpropagation path between the objective function and the backbone network can be changed according to the feature fusion. Since deep neural networks suffer from the gradient vanishing problem, it is difficult to effectively supervise the shallow layer of the backbone network by backpropagation. In contrast, in the feature fusion-based detection framework, all backbone features are directly under the supervision of the objective function. Since this strategy avoids the problem of shallow gradient vanishing, the parameters of each layer of the backbone network can be trained, thus effectively improving the performance of object detection. To further improve the effect, we add a detection head to the original one, so that each layer of the network can get more supervision to train its own parameters.

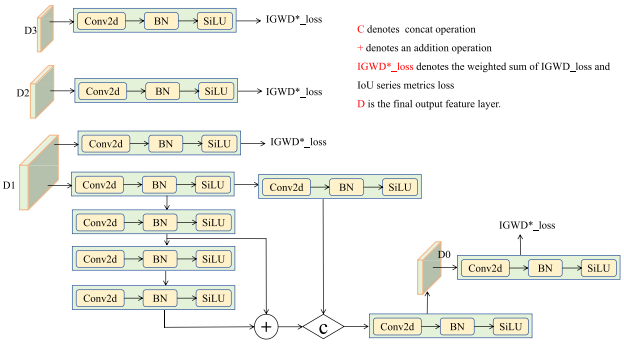


Fig. 4. Model architecture of detection head. D_1, D_2, and D_3 are three feature layers output after FSM processing.

E. Single-Shot Detector: CFENet

The poolFormer module is able to extract global contextual information and reduce the size of the feature map while maintaining spatial information. Using the poolFormer module in the backbone network, the context information around the object can be effectively captured when extracting features to assist in the detection of tiny objects, and effective context information is very important for the detection of tiny objects. As shown in Fig. 1, we use the backbone network built by poolFormer. Then from the base feature layers extracted from the backbone network, we select three different scales of feature layers C_1', C_2', and C_3', which can ensure that the detector can effectively detect objects of different sizes. Typically, the lower layers of features are more useful for detecting larger objects, while the higher layers are more useful for detecting smaller objects. By combining features at multiple scales, you can improve the detector's performance on objects of all sizes.

However, deep features usually contain more semantic information, while shallow features contain more detailed information, which is not effective if detected directly. Therefore, we perform PAN feature fusion on three different extracted feature layers to obtain three feature layers P_1, P_2, and P_3 with rich semantic and detail information at different scales. By combining shallow and deep information, the object detection model can utilize the detailed information for precise localization, while combining semantic understanding for accurate object classification and recognition.

However, while feature fusion methods can effectively improve object detection performance, they also pose challenges, especially when detecting tiny objects. Feature fusion may lead to the feature mixing problem, which makes the features of different scales mixed into the same feature map, resulting in the features of tiny objects being easily covered by the features of large objects and the background information, thus reducing the detection performance. In order to alleviate the problem, we input the feature layers P_1, P_2, and P_3 obtained by feature fusion into the FSM, and obtain the new feature layers D_1, D_2, and D_3. Through the fusion of the FSM mechanism, not only does it alleviate the problem, but also effectively suppresses the background information and enhances the features of the object, and it also retains the advantages of existing feature fusion methods, which further improves the detection performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:07:24 UTC from IEEE Xplore. Restrictions apply.

In addition, to further enhance the detection performance of tiny objects, we perform a convolution operation on top of the feature layer D_1 to obtain the feature layer D_0, which serves as an additional detection head for the shallow feature layer. By adding D_0, the detection performance of tiny objects is effectively improved. Meanwhile, to be able to match more correct prediction boxes and to solve the problem of IoU series loss being sensitive to the positional offset of tiny objects, we use L_{NIG^*} as the loss function of the model, which effectively improves the tiny object detection performance.

IV. Experiments

A. Experiment Setting

1) Dataset: Our proposed method is evaluated on the publicly available datasets AI-TOD [2], [34], VisDrone [35], and DOTA [14].

AI-TOD is a challenging counting set for tiny object detection in aerial images. It has 700 621 object instances and 28 036 aerial images. There are eight object classes: airplane (AI), bridge (BR), storage tank (ST), ship (SH), swimming pool (SP), vehicle (VE), person (PE), and windmill (WM). The average absolute size of AI-TOD is only 12.8 pixels, which is much smaller than that of PASCAL VOC [36] (156.6 pixels), MS COCO [1] (99.5 pixels), and other object detection datasets. AI-TOD defines object sizes (pixels) in the ranges of $[2, 8]$, $[8, 16]$, $[16, 32]$, and $[32, 64]$ as very tiny, tiny, small, and medium, respectively. In the ablation experiments, 11 414 images from the training set are used for training and validation using 2804 images from the validation set. To obtain the final performance compared to the existing methods, we also trained and evaluated the final model on the training set and test set, respectively.

VisDrone contains 10 209 static images (6471 for training, 548 for validation, and 3190 for testing) captured by different drone-mounted cameras and covering a wide range of fields, including location, environment, object, and density. The images were annotated with over 2 600 000 bounding boxes or points of objects of interest. The dataset has ten object categories: car, pedestrian, bicycle, tricycle, motor, people, bus, truck, van, and awning-tricycle. The resolutions of images in the VisDrone range from 540×960 to 2000×1500 pixels. Since the annotation files of the testing set are not public, the validation dataset was used as a test dataset to evaluate the methods.

DOTA is comprised of satellite images. The images in this dataset have a resolution ranging from 800×800 to 4000×4000. Around $280k$ annotated instances are present in the dataset. The objects are from fifteen different categories, with movable objects such as planes, ships, large vehicles, small vehicles, and helicopters. The remaining ten categories are roundabouts, harbors, SPs, etc. Many density crop-based detection papers report results only on movable objects with the assumption that immovable objects usually won't appear crowded. But they are also small objects, so we kept all classes to assess the improvement in small object detection. The training and validation data contain 1411 images and 458 images, respectively.

TABLE I
Comparison of PoolFormer With Other Backbone Networks on the AI-TOD Validation Set

Method	AP	AP_t	params	FLOPs
PoolFormer [15]	**23.5**	**26.4**	15.0	35.8
EMO [41]	19.3	21.8	5.3	50.1
EfficientFormerV2 [42]	22.1	25.7	15.4	26.7
EfficientViT [43]	23.3	26.2	18.4	30.9
ResNet50 [44]	21.9	26.1	30.9	77.7

2) Implementation Details: Although the existing detectors are YOLOv8 [37], YOLOv9 [38], YOLOv10 [39], and YOLO11 [40], YOLOv5 is not weaker than the other new detectors in terms of detection performance of tiny objects, and YOLOv5 has the least number of parameters and computation under the same model architecture. Considering the accuracy and efficiency, we finally chose the COCO pretrained YOLOv5s as representative baseline models. In the following experiments, all methods are implemented on NVIDIA RTX 3090Ti GPU using the Python pytorch framework. The batch size is set to 16 for 200 epochs and the default parameter values of YOLOv5 are used for other parameters. We adopt the same evaluation metric as AI-TOD dataset, including AP, $AP_{0.5}$, $AP_{0.75}$, AP_{vt}, AP_t, AP_s, and AP_m. Specifically, AP is averaged mAP across different IoU thresholds IoU$=\{0.5, 0.55, \cdots, 0.95\}$, $AP_{0.5}$, and $AP_{0.75}$ are AP_s at IoU threshold of 0.5 and 0.75, respectively. In addition, AP_{vt}, AP_t, AP_s, and AP_m are for *very tiny*, *tiny*, *small*, and *medium* scale evaluation in AI-TOD.

B. Ablation Studies

For the ablation experiments, we use COCO pre-trained YOLOv5s as a representative baseline model. The batch size is set to 16 for 200 epochs, and default parameter values of YOLOv5s are used for other parameters. Next, we will perform extensive ablation experiments on the tiny object dataset AI-TOD.

1) Comparison of Different Backbones: First, we compare different backbone networks on the AI-TOD validation set, as shown in Table I. From the table, we can observe that the poolFormer backbone network outperforms other backbone networks for tiny object performance. Therefore, we chose poolFormer as the backbone of CFENet.

2) Ablation Experiments With Single and Multiple Modules: For poolFormer, to effectively utilize the context information around the object to assist the detector model for object detection, we replace the backbone of the baseline with the backbone built by the poolFormer module. as shown in Table II, the detection performance is effectively improved. Using poolFormer can bring 3.0% AP_t improvement for tiny objects and 1.9% AP_{vt} improvement for very tiny objects. This result illustrates that effective and sufficient utilization of the contextual information around the object is also very important for the detection of tiny objects. For FSM, compared with baseline, when CFENet is embedded in FSM, the detector performance is substantially improved by 3.6% AP. This is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:07:24 UTC from IEEE Xplore. Restrictions apply.

TABLE II

ABLATION EXPERIMENTS FOR SINGLE MODULE AND MULTIPLE MODULES ON THE AI-TOD VALIDATION SET (IDH DENOTES IMPROVED DETECTION HEAD, NIG DENOTES IGWD LOSS)

Method	NIG	IDH	FSM	AP	AP_{vt}	AP_t
YOLOv5s				20.6	8.1	23.4
CFENet	-	-	-	23.5	10.0	26.4
	✓	-	-	24.0	7.6	27.3
	-	✓	-	24.0	12.1	27.8
	-	-	✓	24.2	13.5	28.0
	✓	✓	-	24.8	12.6	29.3
	-	✓	✓	25.6	9.4	27.9
	✓	-	✓	25.5	13.8	28.8
	✓	✓	✓	**26.0**	10.9	**29.8**

TABLE III

FSM GENERALIZATION ABILITY TEST ON AI-TOD VALIDATION SET (– INDICATES NO FSM EMBEDDED, P DENOTES PARAMS, F DENOTES FLOPS)

Method	FSM	AP	$AP_{0.5}$	AP_t	AP_s	P	F
PAN	-	20.6	53.3	23.4	25.4	7.2	16.5
	+FSM	**22.1**	**56.5**	**25.5**	**26.1**	7.6	19.2
FPN	-	17.7	47.9	20.8	21.2	4.8	13.2
	+FSM	**19.4**	**51.9**	**22.7**	**23.7**	5.4	16.4
YOLOv7	-	25.4	60.8	28.8	32.2	37.3	105.2
	+FSM	**25.9**	**61.9**	**30.0**	**32.5**	38.0	106.5
YOLOv8s	-	23.8	53.5	23.0	31.7	11.2	28.7
	+FSM	**24.9**	**56.4**	**25.8**	**31.8**	12.0	33.0
YOLOv9s	-	21.1	50.6	21.9	30.2	7.2	27.0
	+FSM	**22.0**	**51.9**	**22.7**	**30.5**	8.3	36.2
YOLOv10s	-	23.3	53.0	24.3	31.3	8.1	24.8
	+FSM	**24.3**	**54.6**	**25.3**	**31.3**	8.9	29.6
YOLO11s	-	23.5	54.1	23.0	30.4	9.4	21.6
	+FSM	**23.8**	**56.7**	**24.6**	**30.9**	10.4	27.2

because our FSM can effectively mitigate the feature mixing problem, suppress the background information interference and redundant features, and thus improve the detection performance of tiny objects. As shown in Table II, we obtain 5.4% AP_{vt} improvement in detecting very tiny objects and 4.6% AP_t improvement in detecting tiny objects, which proves the effectiveness of FSM.

For the IGWD loss function, to optimize the proposed detection network and make it more suitable for tiny object detection, we make some changes to the loss function. Specifically, we replace the original loss with a weighted sum of the IGWD loss and the IoU series loss. The new loss function mitigates the sensitivity of the IoU series loss to the positional offset of tiny objects to some extent and effectively improves the detection performance of tiny objects. The experimental results are shown in Table II. When CFENet uses the IGWD loss function, we find that the performance of AP is further improved by 3.4%, whereas the performance of tiny objects is improved by 3.9%.

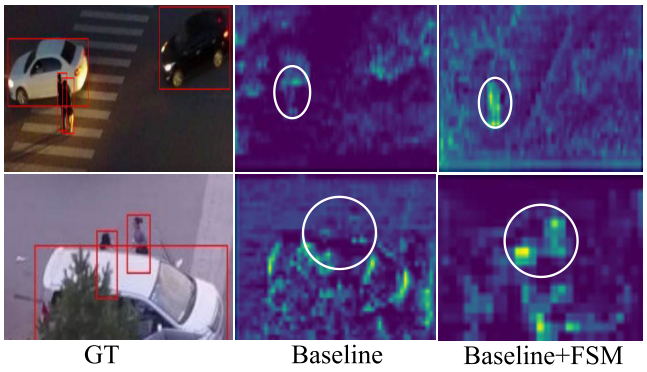


Fig. 5. Comparison results of shallow feature maps. The shallow feature maps were obtained after feeding the images into the baseline (YOLOv5s) and our method. We can see that our method can effectively mitigate the feature mixing problem, suppress the background information, and enhance the features of tiny objects.

TABLE IV

NIG GENERALIZATION ABILITY TEST ON AI-TOD VALIDATION SET (– INDICATES NO NIG EMBEDDED, NIG DENOTES IGWD LOSS)

Method	NIG	AP	$AP_{0.5}$	AP_t	AP_s
YOLOv5s	-	20.6	53.3	23.4	25.4
	+NIG	**21.6**	**53.8**	**24.3**	**27.1**
YOLOv7	-	25.4	60.8	28.8	32.2
	+NIG	**26.1**	**62.2**	**29.9**	**33.4**
YOLOv8s	-	23.8	53.5	23.0	31.7
	+NIG	**24.5**	**58.3**	**25.7**	31.4
YOLOv10s	-	23.3	53.0	24.3	31.3
	+NIG	**24.2**	**55.5**	**25.0**	**32.0**
YOLO11s	-	23.5	54.1	23.0	30.4
	+NIG	**24.4**	**55.5**	**24.9**	**31.3**

For IDH, to further improve the detection performance of tiny objects, we further manipulate the output shallow feature layer D_1 to obtain the feature layer D_0, and then add a detection head on the feature layer D_0. From Table II, it can be seen that adding additional detection heads to the CFENet shallow feature layer can effectively improve the object detection performance. Compared with the baseline, AP_t is improved by 4.4% for tiny objects.

Next, we test the performance variation of fusion between multiple modules. The experimental results are shown in Table II, which shows that when the backbone built with poolFormer is fused with FSM, the AP performance improves by 0.7% and the tiny object performance improves by 1.6%. Further adding a new loss function improves AP by 1.3% and tiny objects by 0.8%. Adding new detection heads to the model resulted in further performance improvements. In the end, the new detection model improved AP by 5.4% and tiny object performance by 6.4% compared to the baseline.

3) FSM, IGWD Loss, and IDH Generalization Ability Test: For FSM, to evaluate the generalization ability of FSM, we improve the YOLOv5s model by keeping only the FPN in the neck part (because the YOLOv5s model neck part is with FPN and PAN), and we call the new model FPN.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:07:24 UTC from IEEE Xplore. Restrictions apply.

TABLE V

IDH GENERALIZATION ABILITY TEST ON AI-TOD VALIDATION SET (– INDICATES NO IDH EMBEDDED)

Method	IDH	AP	$AP_{0.5}$	AP_t	AP_s	AP_m
YOLOv5s	-	20.6	53.3	23.4	25.4	26.6
	+IDH	**22.2**	**24.9**	**25.2**	**27.6**	**30.6**
YOLOv8s	-	23.8	53.5	23.0	31.7	41.3
	+IDH	**24.2**	**54.7**	**24.3**	30.4	**42.1**

TABLE VI

PERFORMANCE COMPARISON ON DIFFERENT LOSSES (NIG DENOTES IGWD LOSS, NIG* DENOTES WEIGHTED SUM OF IGWD LOSS, AND IoU SERIES LOSS)

Metric	YOLOv5s			YOLOv8s		
	AP	$AP_{0.5}$	AP_t	AP	$AP_{0.5}$	AP_t
DIoU	20.1	52.4	22.8	24.2	55.9	24.1
CIoU	20.6	53.3	23.4	23.8	53.5	23.0
GIoU	20.8	53.6	23.1	23.3	53.7	23.2
IoU	20.9	54.5	23.3	20.9	54.5	23.3
EIoU	21.9	54.7	23.9	23.4	53.8	24.1
SIoU	20.5	54.4	24.6	23.9	55.9	25.2
DotD	2.2	9.4	2.4	23.5	55.6	22.3
NWD	10.3	32.1	10.4	22.3	54.6	22.4
KLD	17.9	49.2	19.9	24.0	56.8	24.7
NIG	**21.5**	**55.2**	**25.2**	**24.5**	**58.3**	**25.7**
NIG*	**22.1**	**56.0**	**24.7**	**25.3**	**57.0**	**26.9**

TABLE VII

PERFORMANCE OF DIFFERENT CONSTANTS λ ON AI-TOD VALIDATION SET (YOLOv8s AS BASELINE)

	AP	$AP_{0.5}$	AP_{vt}	AP_t	AP_s
Baseline	23.8	53.5	8.1	23.0	31.7
λ					
1	24.1	56.2	11.5	24.7	30.7
8	24.7	58.3	13.0	24.8	31.6
10	24.5	58.3	10.3	25.7	31.4
13	24.1	57.3	10.5	24.2	31.3

TABLE VIII

IMPROVED DETECTION HEAD GENERALIZATION ABILITY TEST ON AI-TOD VALIDATION SET (0 DENOTES BASELINE. 1 DENOTES ADDING A DETECTOR HEAD TO SHALLOW FEATURE LAYER. 2 DENOTES ADDING A DETECTOR HEAD TO MIDDLE FEATURE LAYER. 3 DENOTES ADDING A DETECTOR HEAD TO DEEP FEATURE LAYER)

DH Pos/Num	AP	$AP_{0.5}$	AP_{vt}	AP_t	AP_s
0	20.6	53.3	8.1	23.4	25.4
1	22.2	54.9	11.2	25.2	27.6
2	21.6	54.7	11.0	25.8	26.2
3	21.6	55.1	10.1	24.3	26.2
1,2	21.6	54.7	12.6	25.8	25.5
1,3	20.7	54.0	10.4	24.7	24.4
2,3	21.4	55.2	11.9	24.6	25.8
1,2,3	21.6	55.4	8.5	25.6	26.1

We apply FSM after FPN and after PAN, respectively, and the experimental results are shown in Table III. From the table, it can be seen that embedding FSM after FPN and PAN, respectively, can effectively improve the detection performance of tiny objects. The experimental results further illustrate that the FSM can effectively alleviate the feature mixing problem. We further test the FSM mechanism on YOLOv7 [61], YOLOv8 [37], YOLOv9 [38], YOLOv10 [39], and YOLO11 [40] models, results are shown in Table III. From the table, we can see that when YOLOv8, YOLOv9, and YOLOv10 are embedded with FSM, object detection results are better than the original model, which also further illustrates the effectiveness and generalization of FSM. As shown in Fig. 5, when baseline embedded FSM can effectively alleviate the feature mixing problem and enhance the features of tiny objects. To further test the generalization ability of the IGWD loss function and IDH, we conducted experiments on different YOLO models. As shown in Tables IV and V, both IGWD loss function and IDH can effectively improve the detection performance of tiny objects.

4) Different Loss Functions: To make the network model more suitable for tiny object detection, we made some modifications to the loss function. Specifically, we replace the

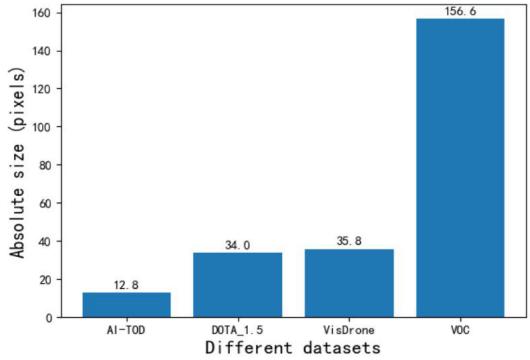


Fig. 6. Comparison of average absolute size of objects in different datasets.

TABLE IX

PERFORMANCE OF DIFFERENT CONSTANTS ON AI-TOD VALIDATION SET

a	AP	$AP_{0.5}$	AP_{vt}	AP_t	AP_s	AP_m
0	20.6	53.3	8.1	23.4	25.4	26.6
0.2	21.0	54.1	11.8	24.2	26.1	29.0
0.3	22.1	56.5	10.4	25.5	26.1	29.1
1	21.4	54.6	10.2	24.8	25.5	27.9

TABLE X

ABLATION EXPERIMENTS FOR SINGLE MODULE AND MULTIPLE MODULES ON DOTA (IDH DENOTES IMPROVED DETECTION HEAD, NIG DENOTES IGWD LOSS)

Method	NIG	IDH	FSM	AP	AP_t	AP_s	AP_m
YOLOv5s	-	-	-	25.3	5.0	9.7	31.9
CFENet	-	-	-	25.5	6.7	10.3	32.1
	✓	-	-	27.2	8.2	11.5	34.3
	-	✓	-	26.0	7.4	10.1	32.4
	-	-	✓	26.9	6.4	10.5	33.6
	✓	✓	-	27.0	5.2	10.2	33.9
	-	✓	✓	27.7	6.8	10.8	35.0
	✓	-	✓	27.6	6.7	11.9	34.5
	✓	✓	✓	27.9	6.2	11.2	35.2

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:07:24 UTC from IEEE Xplore. Restrictions apply.

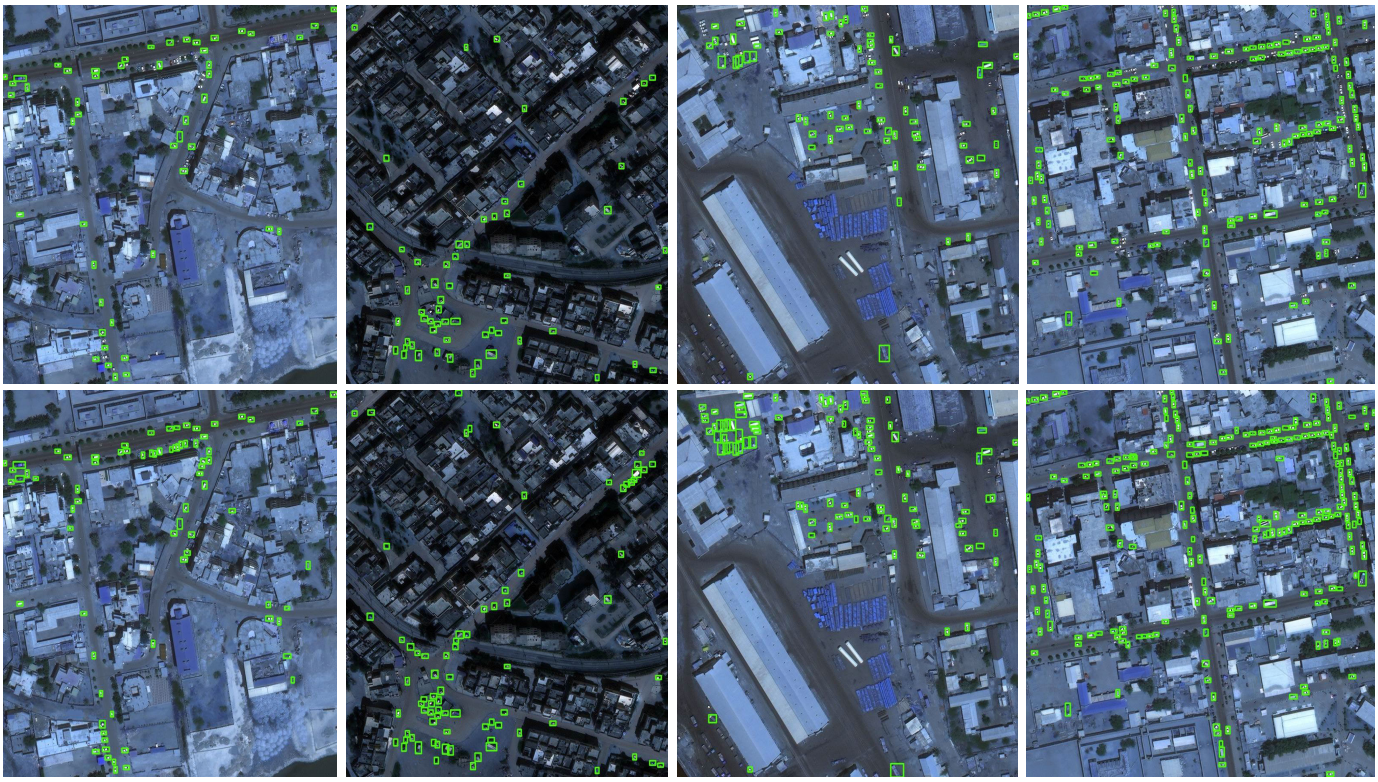


Fig. 7. Visualization of detection results on AI-TOD. The first line is the detection result of the baseline and the second line is the detection result of CFENet. It can be seen that our model is better than the detection result of the baseline.

TABLE XI
FSM AND NIG GENERALIZATION ABILITY TEST ON DOTA
(– INDICATES NO FSM AND NIG EMBEDDED)

Method	NIG	FSM	AP	$AP_{0.5}$	AP_t	AP_m
YOLOv5s	-	-	25.3	41.7	5.0	31.9
	+NIG	-	**26.5**	**41.9**	**7.8**	**33.5**
	-	+FSM	**25.7**	**41.9**	**8.0**	**32.1**
YOLOv8s	-	-	31.2	44.9	9.5	38.5
	+NIG	-	**32.3**	**46.4**	**9.8**	**39.7**
	-	+FSM	**31.8**	**46.1**	**9.6**	**39.1**
YOLOv10s	-	-	32.4	45.9	9.8	40.1
	+NIG	-	**33.1**	**47.1**	**10.4**	**41.0**
	-	+FSM	**32.9**	**46.9**	**10.3**	**40.6**

TABLE XII
ABLATION EXPERIMENTS FOR SINGLE MODULE AND MULTIPLE MODULES ON VISDRONE

Method	AP	$AP_{0.5}$	AP_{vt}	AP_t	AP_s	AP_m
Baseline	14.1	26.2	0.9	4.5	10.8	22.5
CFENet	**17.6**	**31.7**	**1.3**	**6.1**	**14.8**	**26.7**

original loss with a weighted sum of the IGWD loss and the IoU series loss. We compared the loss function with other loss functions based on YOLOv5s and YOLOv8s, and the experimental results are shown in Table VI. From the table, it can be seen that when using the new loss function, the performance of AP and AP_t is better than the other loss functions, which

TABLE XIII
FSM AND NIG GENERALIZATION ABILITY TEST ON VISDRONE
(– INDICATES NO FSM AND NIG EMBEDDED)

Method	NIG	FSM	AP	$AP_{0.5}$	AP_t	AP_s	AP_m
YOLOv10s	-	-	18.2	31.1	5.8	15.2	28.5
	+NIG	-	**19.1**	**32.2**	**6.0**	**15.6**	**29.8**
	-	+FSM	**19.0**	**32.0**	**6.1**	**15.9**	**29.3**

further indicates that the new loss function is more suitable for the detection of tiny objects. Next, we discuss the effect of different values of the hyperparameter λ on the detection performance using the loss function as an example, and the results are shown in Table VII. As can be seen from the table, when λ takes the value of 8, the value of AP reaches the highest and improves 0.9% over the benchmark value. From the results in the table, it can also be concluded that the hyperparameter λ has an effect on the detection results, so in our future work, we will investigate the adaptive hyperparameter to further improve the performance. Overall the new metrics outperform the baseline in detecting tiny objects.

5) Comparison of Multiple Detection Heads: We next tested the change in model performance when adding detection heads on different output feature layers. The results of the comparison experiments of adding additional detection heads for different feature layers separately are shown in Table VIII. From the table, we can observe that when a detection head is added to the output feature layer for further operations in the shallow feature layer, the AP performance is better than the other feature layers, with an increase of 1.6% in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:07:24 UTC from IEEE Xplore. Restrictions apply.

TABLE XIV
PERFORMANCE OF STATE-OF-THE-ART DETECTORS ON THE PROPOSED AI-TOD TEST SET (* DENOTES THAT THE MODEL IS EMBEDDED IN OUR METHOD)

Method	Backbone	AP	$AP_{0.5}$	$AP_{0.75}$	AP_{vt}	AP_t	AP_s	AP_m
anchor-based								
SSD-512 [45]	ResNet-50	7.0	21.7	2.8	1.0	4.7	11.5	13.5
TridentNet [46]	ResNet-50	7.5	20.9	3.6	1.0	5.8	12.6	14.0
CenterNet [47]	DLA-34	13.4	39.2	5.0	3.8	12.1	17.7	18.9
DotD [14]	ResNet-50	16.1	39.2	10.6	8.3	17.6	18.1	22.1
ATSS [48]	ResNet-50	12.8	30.6	8.5	1.9	11.6	19.5	29.2
RetinaNet [49]	ResNet-50	4.7	13.6	2.1	2.0	5.4	6.3	7.6
Faster R-CNN [50]	ResNet-50	11.1	26.3	7.6	0.0	7.2	23.3	33.6
Cascade R-CNN [51]	ResNet-50	13.8	30.8	10.5	0.0	10.6	25.5	36.6
DetectoRS [20]	ResNet-50	14.8	32.8	11.4	0.0	10.8	28.3	38.0
PAA [52]	ResNet-50	10.0	26.5	6.7	3.5	10.5	13.1	22.1
OTA [53]	ResNet-50	10.4	24.3	7.2	2.5	11.9	15.7	20.9
NWD [7]	ResNet-50	20.8	49.3	14.3	6.4	19.7	29.6	38.3
NWD-RKA [34]	ResNet-50	24.7	57.4	17.1	9.7	24.2	29.8	**39.3**
RFLA [54]	ResNet-50	24.8	55.2	18.5	9.3	24.8	30.3	38.2
YOLOv5 [32]	CSPDarkNet-s	22.3	55.9	13.7	7.5	23.3	26.7	28.5
HANet [3]	ResNet-50	22.1	53.7	14.4	10.9	22.2	27.3	36.8
DNTR [55]	ResNet-50	26.2	56.7	20.2	**12.8**	26.4	31.0	37.0
HS-FPN [56]	ResNet-50	25.1	55.7	19.1	12.1	25.3	29.9	36.9
anchor-free								
FoveaBox [57]	ResNet-50	8.1	19.8	5.1	0.9	5.8	13.4	15.9
PepPonits [58]	ResNet-50	9.2	23.6	5.3	2.5	9.2	12.9	14.4
FCOS [59]	ResNet-50	9.8	24.1	5.9	1.4	8.0	15.1	17.4
Grid R-CNN [60]	ResNet-50	12.2	27.7	9.0	0.2	10.3	22.6	23.3
YOLOv8 [37]	DarkNet-s	24.5	55.0	17.9	7.1	23.7	32.0	37.3
YOLOv8*	DarkNet-s	**25.4**	**56.6**	**19.2**	**8.0**	**25.3**	**32.5**	**38.4**
YOLOv10 [39]	CSPNet-s	24.7	54.7	18.7	7.6	24.7	32.2	36.9
YOLOv10*	CSPNet-s	**26.1**	**56.5**	**20.2**	**7.9**	**26.1**	**33.4**	**38.7**
YOLO11 [40]	C3k2Net-s	25.7	56.6	19.7	7.3	25.7	33.8	38.6
YOLO11*	C3k2Net-s	**26.2**	**57.4**	**19.8**	**7.5**	**26.3**	**34.0**	**39.7**
FFCA-YOLO [18]	CSPDarkNet53	24.7	**59.5**	15.7	9.7	26.0	28.4	29.5
CFENet	CSPDarkNet-s	**26.6**	59.2	**19.6**	9.0	**27.5**	**32.3**	38.5

AP value and 1.8% in the performance of the tiny objects as compared to the baseline. It also further illustrates that a certain number of detection heads has a certain impact on the detection performance.

6) Hyperparameter: Next, the effect of different values of hyperparameter a on the detection performance is discussed and the results are shown in Table IX. From the table, it can be observed that different values of a have some effect on the detection performance. When a is taken as 0.3, the AP value reaches the highest and increases by 1.5% over the baseline, and AP_t is improved by 2.1% for tiny objects. From the results in the table, it can also be concluded that the hyperparameter a has a certain effect on the detection results, so in future work, we will study the adaptive hyperparameters to further improve the performance.

7) Ablation Experiments on DOTA and VisDrone Datasets: To further validate the generalization ability of our proposed method, we have done extensive experiments on the public datasets DOTA and VisDrone. As shown in Tables X–XIII, our proposed model and method have good generalization ability and outperform other models. It is worth noting that the average absolute size of the VisDrone and DOTA datasets is much larger than that of the AI-TOD dataset as shown in Fig. 6, whereas the CFENet model and the other methods are designed for tiny objects, most of the objects in the DOTA and VisDrone datasets are concentrated on medium objects, and the evaluation metrics that we use are for the AI-TOD so that our proposed method has limited improvement on tiny objects on DOTA and VisDrone, but significant improvement on small and medium objects.

C. Main Results on AI-TOD

To validate the effectiveness of the CFENet model, we conducted experiments on the tiny object detection dataset AI-TOD and compared it with the current state-of-the-art methods. As shown in Table XIV, from the experimental results, we can see that our CFENet detection performance is better than the state-of-the-art detectors, especially on tiny

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:07:24 UTC from IEEE Xplore. Restrictions apply.

objects, which is 4.2% higher compared to the baseline and 1.1% higher compared to the state-of-the-art detectors. This further indicates that CFENet can effectively utilize the surrounding context information, suppress redundant features and background information, enhance the features of tiny objects, and improve the detection performance of tiny objects. As shown in Fig. 7, we visualize the detection results, from which we can observe that our method outperforms the baseline.

V. Conclusion

In this article, a CFENet model is proposed for tiny object detection. In the CFENet model, to effectively utilize the context information around the tiny objects to assist the tiny objects for detection, we build the backbone of the CFENet model with the poolFormer module. To alleviate the feature mixing problem caused by feature fusion, suppress the background information and object redundant features, and enhance the tiny object features, we propose the FSM mechanism, which further improves the tiny object detection performance by embedding it after the feature fusion operation in the CFENet model. Since the loss function of the IoU family of metrics is very sensitive to the positional offset of tiny objects, to alleviate this problem, we improve the loss function and propose the IGWD loss function, which effectively improves the detection performance of tiny objects. In addition, to further improve the tiny object detection performance, we further manipulate the output shallow feature layer and add additional detection heads. The effectiveness of our method is further demonstrated by experiments on the AI-TOD, VisDrone, and DOTA datasets, and the experimental results outperform existing detection methods.

References

[1] T. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, Zurich, Switzerland. Cham, Switzerland: Springer, Sep. 2014, pp. 740–755.

[2] J. Wang, W. Yang, H. Guo, R. Zhang, and G.-S. Xia, "Tiny object detection in aerial images," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 3791–3798.

[3] G. Guo, P. Chen, X. Yu, Z. Han, Q. Ye, and S. Gao, "Save the tiny, save the all: Hierarchical activation network for tiny object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 1, pp. 221–234, Jan. 2023.

[4] W. Ma, X. Wang, H. Zhu, X. Yang, X. Yi, and L. Jiao, "Significant feature elimination and sample assessment for remote sensing small objects' detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[5] B. Singh, M. Najibi, and L. S. Davis, "SNIPER: Efficient multi-scale training," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2018, pp. 9310–9320.

[6] X. Yu, Y. Gong, N. Jiang, Q. Ye, and Z. Han, "Scale match for tiny person detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, Jun. 2020, pp. 1257–1265.

[7] J. Wang, C. Xu, W. Yang, and L. Yu, "A normalized Gaussian Wasserstein distance for tiny object detection," 2021, *arXiv:2110.13389*.

[8] T. Kong, F. Sun, A. Yao, H. Liu, M. Lu, and Y. Chen, "Ron: Reverse connection with objectness prior networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2017, pp. 5936–5944.

[9] Z. Zhang, S. Qiao, C. Xie, W. Shen, B. Wang, and A. L. Yuille, "Single-shot object detection with enriched semantics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5813–5821.

[10] T. Kong, A. Yao, Y. Chen, and F. Sun, "HyperNet: Towards accurate region proposal generation and joint object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 845–853.

[11] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[12] C. Yang, Z. Huang, and N. Wang, "QueryDet: Cascaded sparse query for accelerating high-resolution small object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13668–13677.

[13] J. Yu, Y. Jiang, Z. Wang, Z. Cao, and T. Huang, "UnitBox: An advanced object detection network," in *Proc. 24th ACM Int. Conf. Multimedia*, Oct. 2016, pp. 516–520.

[14] C. Xu, J. Wang, W. Yang, and L. Yu, "Dot distance for tiny object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 1192–1201.

[15] W. Yu et al., "MetaFormer is actually what you need for vision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10819–10829.

[16] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[17] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10781–10790.

[18] Y. Zhang, M. Ye, G. Zhu, Y. Liu, P. Guo, and J. Yan, "FFCA-YOLO for small object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[19] W. Ma et al., "Feature split–merge–enhancement network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[20] S. Qiao, L.-C. Chen, and A. Yuille, "DetectoRS: Detecting objects with recursive feature pyramid and switchable atrous convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10213–10224.

[21] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 658–666.

[22] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren, "Distance-IoU loss: Faster and better learning for bounding box regression," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2020, vol. 34, no. 7, pp. 12993–13000.

[23] Y.-F. Zhang, W. Ren, Z. Zhang, Z. Jia, L. Wang, and T. Tan, "Focal and efficient IOU loss for accurate bounding box regression," *Neurocomputing*, vol. 506, pp. 146–157, Sep. 2022.

[24] Z. Gevorgyan, "SIoU loss: More powerful learning for bounding box regression," 2022, *arXiv:2205.12740*.

[25] C. Chen, M.-Y. Liu, O. Tuzel, and J. Xiao, "R-CNN for small object detection," in *Proc. 13th Asian Conf. Comput. Vis. (ACCV)*, Taipei, Taiwan. Cham, Switzerland: Springer, Jan. 2017, pp. 214–230.

[26] X. Tang, D. K. Du, Z. He, and J. Liu, "PyramidBox: A context-assisted single shot face detector," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 797–813.

[27] X. Liang, J. Zhang, L. Zhuo, Y. Li, and Q. Tian, "Small object detection in unmanned aerial vehicle images using feature fusion and scaling-based single shot detector with spatial context analysis," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 6, pp. 1758–1770, Jun. 2020.

[28] X. Hu et al., "SINet: A scale-insensitive convolutional neural network for fast vehicle detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 3, pp. 1010–1019, Mar. 2019.

[29] S. Bell, C. L. Zitnick, K. Bala, and R. Girshick, "Inside-outside net: Detecting objects in context with skip pooling and recurrent neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2874–2883.

[30] T. Zhang et al., "Multistage enhancement network for tiny object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[31] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[32] G. Jocher et al. (2022). *Ultralytics/yolov5: V7. 0*. [Online]. Available: https://github.com/ultralytics/yolov5/tree/v7.0

[33] X. Yang, J. Yan, Q. Ming, W. Wang, X. Zhang, and Q. Tian, "Rethinking rotated object detection with Gaussian Wasserstein distance loss," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, vol. 139, Jul. 2021, pp. 11830–11841.

[34] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "Detecting tiny objects in aerial images: A normalized Wasserstein distance and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 79–93, Aug. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:07:24 UTC from IEEE Xplore. Restrictions apply.

[35] D. Du et al., "VisDrone-DET2019: The vision meets drone object detection in image challenge results," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 213–226.

[36] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, 2010.

[37] G. Jocher, A. Chaurasia, and J. Qiu. (2023). *YOLO by Ultralytics*. [Online]. Available: https://github.com/ultralytics/ultralytics

[38] C.-Y. Wang, I.-H. Yeh, and H.-Y. Mark Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," 2024, *arXiv:2402.13616*.

[39] A. Wang et al., "YOLOv10: Real-time end-to-end object detection," 2024, *arXiv:2405.14458*.

[40] G. Jocher, A. Chaurasia, and J. Qiu. (2024). *YOLO by Ultralytics*. [Online]. Available: https://github.com/ultralytics/ultralytics

[41] J. Zhang et al., "Rethinking mobile block for efficient attention-based models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 1389–1400.

[42] Y. Li et al., "Rethinking vision transformers for mobilenet size and speed," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 16889–16900.

[43] X. Liu, H. Peng, N. Zheng, Y. Yang, H. Hu, and Y. Yuan, "EfficientViT: Memory efficient vision transformer with cascaded group attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 14420–14430.

[44] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[45] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. 14th Eur. Conf. Comput. Vis.*, Amsterdam, The Netherlands. Cham, Switzerland: Springer, Oct. 2016, pp. 21–37.

[46] Y. Li, Y. Chen, N. Wang, and Z.-X. Zhang, "Scale-aware trident networks for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6054–6063.

[47] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6569–6578.

[48] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9759–9768.

[49] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[50] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Int. Conf. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.

[51] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[52] K. Kim and H. S. Lee, "Probabilistic anchor assignment with IoU prediction for object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K. Cham, Switzerland: Springer, Aug. 2020, pp. 355–371.

[53] Z. Ge, S. Liu, Z. Li, O. Yoshie, and J. Sun, "OTA: Optimal transport assignment for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 303–312.

[54] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "RFLA: Gaussian receptive field based label assignment for tiny object detection," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 526–543.

[55] H.-I. Liu, Y.-W. Tseng, K.-C. Chang, P.-J. Wang, H.-H. Shuai, and W.-H. Cheng, "A DeNoising FPN with transformer R-CNN for tiny object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4704415.

[56] Z. Shi et al., "HS-FPN: High frequency and spatial perception FPN for tiny object detection," 2024, *arXiv:2412.10116*.

[57] T. Kong, F. Sun, H. Liu, Y. Jiang, L. Li, and J. Shi, "FoveaBox: Beyound anchor-based object detection," *IEEE Trans. Image Process.*, vol. 29, pp. 7389–7398, 2020.

[58] Z. Yang, S. Liu, H. Hu, L. Wang, and S. Lin, "RepPoints: Point set representation for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9657–9666.

[59] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9627–9636.

[60] X. Lu, B. Li, Y. Yue, Q. Li, and J. Yan, "Grid R-CNN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7363–7372.

[61] C.-Y. Wang, A. Bochkovskiy, and H.-Y.-M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7464–7475.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:07:24 UTC from IEEE Xplore. Restrictions apply.